

quantmsdiann: a scalable SDRF-driven DIA-NN workflow for reanalysis of single-cell, spatial, and bulk proteomics datasets

Qi-Xuan Yue^a, Yufei Shen^a, Asier Larrea^c, Chengxin Dai^b, Henry Webel^d, Jose Nimo^f, Fabian Coscia^f, Orhun Kok^g, Nikolai Slavov^{g,h}, Timo Sachsenberg^{i,j}, Mingze Bai^a, Vadim Demichev^e, Yasset Perez-Riverol^{c,*}

^aChongqing Key Laboratory of Big Data for Bio Intelligence, Chongqing University of Posts and Telecommunications, Chongqing, China

^bState Key Laboratory of Proteomics, Beijing Proteome Research Center, National Center for Protein Sciences (Beijing), Beijing Institute of Life Omics, Beijing, China

^cEuropean Molecular Biology Laboratory, European Bioinformatics Institute (EMBL-EBI), Wellcome Genome Campus, Hinxton, Cambridge, UK

^dNovo Nordisk Foundation Center for Protein Research, Faculty of Health and Medical Sciences, University of Copenhagen, Copenhagen, Denmark

^eQuantitative Proteomics laboratory, Charité – Universitätsmedizin Berlin, Berlin, Germany

^fSpatial Proteomics Group, Max-Delbrück-Center for Molecular Medicine in the Helmholtz Association (MDC), Berlin, Germany

^gDepartment of Bioengineering, Department of Biology, Department of Chemistry and Chemical Biology, Single Cell Proteomics Center, and Barnett Institute for Chemical and Biological Analysis, Northeastern University, Boston, Massachusetts, USA

^hParallel Squared Technology Institute, Watertown, Massachusetts, USA

ⁱDepartment of Computer Science, Applied Bioinformatics, University of Tübingen, Tübingen, Germany

^jInstitute for Bioinformatics and Medical Informatics, University of Tübingen, Tübingen, Germany

Abstract

Public proteomics archives now host thousands of data-independent acquisition (DIA) datasets, but they remain difficult to integrate and reuse because each dataset has been processed with a different software configuration, and most of them lack standardised metadata. Here, we present **quantmsdiann**, an open-source Nextflow/nf-core workflow that parallelises DIA-NN across cloud and HPC infrastructures, guided by the experimental design declared in SDRF format. The workflow ships container-pinned profiles for each supported DIA-NN version (extensible to any release via a custom container) through Docker and Singularity, with no manual installation or dependency resolution, and natively handles all vendor formats, converting to mzML when needed (e.g. SCIEX). It exports harmonised quantification tables as MSstats input, the new Quantitative Proteomics eXchange format (QPX), and a **pmultiqc** quality-control report. The parallelisation design reanalyses a 2,300-run single-cell dataset in 2.2 hours on an HPC cluster of 300 nodes. We benchmark **quantmsdiann** on four ProteoBench DIA instrument modalities, where distributed **quantmsdiann** reproduces single-machine community submissions independent of DIA-NN release. We reanalyse public DIA deposits spanning single-cell proteomes (HeLa on Orbitrap Astral, PXD046357; One-Tip, PXD044991; and an oocyte plexDIA cohort, MSV000093870) and bulk cell-line panels (NCI-60/PXD003539 and ProCan/PXD030304, with three further cohorts combined into a five-cohort pan-cohort under a single invocation). Reprocessing with current DIA-NN

*Corresponding author.

URL: yasset@ebi.ac.uk (Yasset Perez-Riverol)

releases matches or exceeds the originally deposited counts: across these single-cell and bulk cell-line cohorts it recovers on average $\sim 22\%$ more protein groups (up to 44% in single cells) and, in the single-cell cohorts, $\sim 32\%$ more precursors, than the original or 1.8.1 analyses; a spatial Deep Visual Proteomics cohort (PXD064049) and a phosphoproteomics cohort (PXD049692) are reanalysed in the supplement. `quantmsdiann` is a step towards scalable, reproducible reanalysis of the growing DIA data landscape and a foundation for community efforts in large-scale DIA meta-analysis and atlas building. `quantmsdiann` is available at <https://github.com/bigbio/quantmsdiann>.

Keywords: DIA-NN, single-cell proteomics, spatial proteomics, phosphoproteomics, plexDIA, data-independent acquisition, Nextflow, nf-core, SDRF, reproducibility, QPX, pmultique

1. Introduction

Data-independent acquisition (DIA) is now the dominant mass-spectrometry strategy for proteomics at scale, and DIA-NN [1] is among the most widely adopted analysis engines across timsTOF [2, 3], Astral, and Orbitrap platforms. Recent benchmarks confirm DIA-NN’s competitiveness for single-cell DIA workflows [4], and recent releases have expanded support for plexDIA, timsTOF parallel accumulation-serial fragmentation (PASEF), and vendor-native raw-file reading. Despite this analytical maturity, most DIA-NN users run the tool as a desktop application, which limits its scalability for large datasets. Deployments on HPC and cloud infrastructure remain difficult, relying on shell scripts, bespoke directory conventions, and hand-curated sample sheets. As a result, the same raw data analysed by two laboratories, or with two DIA-NN releases, can yield non-overlapping quantification tables that are difficult to reconcile.

Beyond per-study analysis, the value of public proteomics archives such as PRIDE Archive [5] and other ProteomeXchange partners [6] lies as much in systematic reanalysis as in original deposition. In 2024 we published `bigbio/quantms` [7], a Nextflow pipeline that standardised multi-engine DIA and DDA analyses behind a Sample and Data Relationship Format (SDRF) [8] input on the `nf-core` framework [9]. `quantms` was, however, built around DIA-NN 1.8.1 with a fixed parameter set, and its DDA/DIA combination within the same workflow made extending DIA-specific functionality difficult. New DIA-NN parameters conflicted with DDA options, versions were hard-coded, sample-level acquisition parameters were not properly propagated to per-run DIA-NN invocations, and raw-file conversion was mandatory. Iterative reanalysis at archive scale requires a workflow that combines the reproducibility of `nf-core`, the sensitivity of current DIA-NN releases, the metadata discipline of SDRF, and a parallelisation strategy capable of processing thousands of raw files.

Here, we present `quantmsdiann`, an independent SDRF-driven Nextflow pipeline dedicated to DIA-NN that extends DIA-specific features beyond what the original `bigbio/quantms` pipeline could accommodate. While `quantmsdiann` shares its SDRF-first design philosophy and parts of the configuration-generation tooling with `quantms`, its workflow, container set, and module library are developed and released independently, with container-pinned profiles for every DIA-NN version selectable at runtime under Docker or Singularity. Downstream, it exports a harmonised QPX (Quantitative Proteomics eXchange) Parquet archive alongside the standard DIA-NN reports and MSstats input. We demonstrate the pipeline along three axes: ProteoBench [10] benchmarks across four DIA modalities and successive DIA-NN releases; independent reanalyses of public DIA deposits spanning bulk cell lines (NCI-60, ProCan), a single-cell datasets, a spatial Deep Visual Proteomics cohort, and a phosphoproteomics cohorts; and a pan-cohort of five public DIA cell-line reanalyses integrated under a single SDRF-driven invocation. Across the single-cell and bulk cell-line cohorts, reanalysis with the current DIA-NN build recovers on average $\sim 22\%$ more protein groups (and $\sim 32\%$

more precursors in single cells) than the original or 1.8.1 analyses, so reanalysing archived data with an up-to-date engine recovers measurably more of each deposit.

2. Experimental Procedures

2.1. Pipeline implementation

`quantmsdiann` is implemented in Nextflow DSL2 ($\geq 25.10.4$) and follows nf-core community standards [9] for pipeline structure, testing, documentation, and continuous integration. It supports both DIA-NN library-free analysis (predicting the spectral library in-silico from a FASTA reference) and library-based analysis against a user-supplied spectral library; all analyses in this study used the library-free mode with the UniProt human reference proteome [11] (UP000005640, downloaded March 2026). Each computational step is packaged as a versioned container image (Docker, Singularity, or Apptainer). Open-source community tools (ThermoRawFileParser [12], pridepy [13], QPX, and the `wiffconverter` SCIEX reader) are pulled from public registries (BioContainers [14] or `ghcr.io/bigbio`). Because the DIA-NN license restricts redistribution, only version 1.8.1 is shipped as an image (the default, from BioContainers); releases from 1.9 onward (1.9.2–2.5.1, including the 2.5.1 enterprise build) are built locally from the per-release recipes in the companion `quantms-containers` repository (<https://github.com/bigbio/quantms-containers>) and tagged so the workflow selects them automatically (Supplementary Note 2). The repository at <https://github.com/bigbio/quantmsdiann> is organised as a standard nf-core pipeline with main workflow (`main.nf`), subworkflows, modules, test data and CI configurations, and documentation. Pipeline version v2.2.0 (release codename “Chongqing”, 2026-06-16) was used for the analyses reported in this paper.

2.2. SDRF extensions for `quantmsdiann`

`quantmsdiann` accepts SDRF [8] as primary input. SDRF parsing is performed using the `sdrf-pipelines` Python package developed within the `quantms` project, which validates the SDRF against the proteomics-specific subset of EFO, PSI-MS, and PRIDE ontology terms and emits a normalised per-sample channel for downstream Nextflow processes. To express DIA- and DIA-NN-specific acquisition settings at the sample level, we extended the SDRF with additional comment columns: the MS1 and MS2 scan mass-window intervals and, for multiplexed experiments, the plexDIA channel definitions and their isotopologue labels. Because these settings are declared per sample, heterogeneous MS1/MS2 window schemes, or a plexDIA channel layout, can be mixed within a single cohort without any change to the workflow. We added a `convert-diann` command to `sdrf-pipelines` that reads these extended columns together with the standard fields (enzyme, fixed and variable modifications, instrument, precursor mass tolerance, fragment mass tolerance, and missed-cleavage settings) and writes a per-run DIA-NN command-line configuration; where SDRF columns are absent, DIA-NN defaults appropriate for the inferred instrument are applied.

2.3. Raw file handling and DIA-NN execution

A key advance over its predecessor `quantms` is that native vendor (raw) formats are first-class inputs in `quantmsdiann`: whereas `quantms` converted every raw file to mzML up front, `quantmsdiann` reads each instrument’s native format directly and converts only when a particular format or DIA-NN version requires it. Thermo `.raw` or `timsTOF` (`.d` or `.d.zip`) files are read natively by DIA-NN $\geq 2.1.0$; for DIA-NN versions lower than 2.1.0 conversion to mzML is performed with ThermoRawFileParser [12] (`-mzml_convert`); Bruker `.d` folders are ingested natively, with compressed forms (`.d.tar`/`.zip`/`.gz`) decompressed first; and Sciex `.wiff` / `.wiff2` are converted

to mzML with the open-source `wiffconverter` (<https://github.com/bigbio/wiffconverter>). Generic `.dia` and pre-converted `.mzML` inputs pass through unchanged, with optional OpenMS re-indexing (`-reindex_mzml`).

2.4. Quality control and QPX export

Per-run and per-cohort quality control reports are generated by `pmultiqc` [15], a proteomics extension of MultiQC [16] that summarises DIA-NN-specific metrics including peptide and protein counts, missed cleavages, charge-state distributions, precursor and fragment mass accuracy, and per-run identification yield. The QPX archive is produced by the `qpxc` CLI from the `qpx` package (<https://github.com/bigbio/qpx>), which writes Parquet files for identifications (psm, peptide, protein-group, feature), metadata (sample, run, ontology), and provenance, and optionally exports MuData (`.h5mu`) containers for scverse-based [17] downstream analysis. The aim of exporting to QPX is to bundle the SDRF, identifications, quantifications, and pipeline provenance into a single queryable artefact for reanalysis and long-term archival, and to provide a standardised input for downstream tools that support the format (e.g. MSstats, Perseus, or custom scripts).

2.5. Datasets benchmarked and compute environment

Supplementary Note 1 (Table S1) lists all benchmark and reanalysis datasets with their instrument, acquisition mode, and deposition year, and an SDRF prepared from the original sample metadata. They fall into four experiment classes: *ProteoBench* DIA benchmarks (modules 5, 7, 9, and 10); *single-cell and spatial proteomics*, comprising the HeLa single-cell cohorts PXD046357 (Orbitrap Astral) and PXD044991 (One-Tip), the oocyte plexDIA cohort MSV000093870, the PXD064049 spatial Deep Visual Proteomics cohort, and the PXD071075 single-cell scalability sweep; *bulk cell-line panels* (PXD003539, PXD030304, PXD004701, PXD041421, and PXD017199); and *phosphoproteomics* (PXD034128, PXD034623, and PXD049692). For each cell-line reanalysis cohort, the original published quantification was downloaded from PRIDE as the comparator, with both it and the `quantmsdiann` reanalysis thresholded at 1% precursor-level FDR; the conservative contaminant/decoy filter and the like-for-like-threshold rule applied to the original headline counts are described in Supplementary Note 3. Runtime benchmarks were run on the EMBL-EBI HPC cluster (LSF scheduler), with per-task CPU and memory dispatched by Nextflow and memory profiled from the Nextflow trace report (`-with-trace`).

3. Results

3.1. `quantmsdiann` pipeline architecture

Built on the `nf-core` framework [9], `quantmsdiann` accepts a single SDRF file as primary input, which encodes per-sample metadata (cell line or tissue origin, factor values, instrument, post-translational modifications) and per-run file references. We extended the SDRF with DIA-relevant columns (MS1 and MS2 mass windows, and plexDIA channel definitions and labels) and added an adapter to `sdrf-pipelines` [8] (`convert-diann`) that translates each run’s declared metadata into a per-run DIA-NN command-line configuration. Every run is therefore parametrised from its own metadata with no manual configuration or parameter-harmonisation step, removing a common source of analysis-script divergence between groups: different MS1/MS2 window schemes can be mixed within a cohort, or a plexDIA dataset run with its channel definitions, without any workflow change beyond the SDRF. The workflow comprises four mandatory and three optional modules organised around the DIA-NN analysis core (Fig. 1a), alternating between per-file parallel stages (red; raw-file preparation, preliminary calibration, and individual search) and collective stages (green;

in-silico library generation, empirical library assembly, final quantification, MSstats conversion, and `pmultiqc` [15] quality-control reporting) operating on the full cohort.

`quantmsdiann` follows DIA-NN’s distributed multi-pass flow, alternating parallel per-file work with serial cohort-wide steps (Fig. 1a). Each raw file is first prepared *in parallel* (vendor-format conversion, decompression, or indexing) where needed; conversion to mzML is mandatory for early DIA-NN versions and for some vendor formats such as SCIEX `.wiff`. A spectral library is then predicted in-silico from the FASTA once (*serial*; skipped when a library is supplied), and every file is calibrated against it in a *parallel* first pass. The first-pass results are merged into a single empirical library (*serial*); each file is searched again against that library *in parallel*; and a final consolidation pass merges all per-file results into the cohort quantification, applying match-between-runs, normalisation, and protein inference (*serial*). MSstats conversion, `pmultiqc` reporting, and QPX export then run once at the end (*serial*). The two per-file search passes carry essentially all of the compute and scale across cluster nodes, driving the throughput reported below (Section 3.2); the serial steps set a largely fixed wall-clock floor. Outputs include the standard DIA-NN report tables: `report.tsv` (or `report.parquet`), `report.pr_matrix.tsv`, and `report.pg_matrix.tsv`, alongside MSstats-formatted input and a `pmultiqc` [15] quality-control report. A harmonised QPX archive bundles identifications, metadata, and provenance into a Parquet-based artefact, queried directly with DuckDB and, for the quantification layers, exported to MuData (`.h5mu`) for scverse-compatible [17] downstream analysis.

We evaluate `quantmsdiann` along four lines that together motivate it as a reanalysis platform. First, because every release is provided as a container-pinned profile selectable at runtime, the same SDRF can be analysed with any supported DIA-NN version (1.8.1 through 2.5.1, including the enterprise build, and any other release via a custom container), so the engine version is a runtime choice rather than a reinstallation. Second, the per-file parallel design lets a single invocation scale with data volume, from six-run benchmarks to cohorts of thousands of single-cell runs (Section 3.2). Third, on the ProteoBench community benchmarks this distributed execution reproduces the identification and quantification performance of single-machine DIA-NN runs, i.e. scaling out does not change the result relative to a desktop analysis (Section 3.3). Fourth, with that equivalence established, reanalysing public deposits with current DIA-NN releases recovers more identifications than the originally deposited analyses, with the largest gains on low-input single-cell data (Sections 3.4–3.5).

3.2. *quantmsdiann* throughput and scaling

To characterise how `quantmsdiann` scales on an HPC cluster or cloud, we simulated the reanalysis of an entire single-cell experiment of ~2,300 raw files (PXD071075) across cluster sizes from 10 to 300 nodes (set through the Nextflow `executor.queueSize`), and measured the end-to-end wall-clock time (Fig. 1b). Because the per-file work is parallelised across nodes, a single SDRF-driven invocation completed the whole cohort within hours: wall-clock fell monotonically from 37.4 h at 10 nodes to 2.2 h at 300 nodes (excluding the one-time in-silico library-prediction step, which is independent of cluster size). The curve has a practical knee near 200 nodes (2.4 h), beyond which the serial cohort-level steps dominate and additional nodes give diminishing returns (a 30× increase in cluster width yields only a 17.4× speed-up), so the cluster should be sized to the desired turn-around time rather than to maximum throughput.

We then compared the time-to-finish across every dataset in this study (Fig. 1c), reporting each run net of the one-time in-silico library-prediction step. We didn’t include the insilico-library generation because it can’t be parallelize and depedent a lot on the type of the paremeter for the library generation (e.g. number of modifications or MS1 and MS2 windows). Across cohorts

spanning three orders of magnitude in file count, wall-clock stayed within hours because the per-file passes absorb additional files in parallel: the 5,798-run ProCan cohort (PXD030304) finished in 6.2 h, in the same band as cohorts orders of magnitude smaller, while the 6-run ProteoBench benchmarks completed in under 1.5 h. Time-to-finish is therefore governed by per-run search cost and cohort-level consolidation rather than by the number of files: the longest reanalysis was a 206-run cohort on an older Q Exactive instrument (PXD017199, 11.4 h), exceeding the far larger ProCan cohort, followed by a 10-file phosphoproteomics cohort whose deep variable-modification search drives its 8.0 h (PXD049692). Peak memory during the cohort-level consolidation stayed within the 64–128 GB available on commodity HPC nodes, so no high-memory specialist hardware is required. Per-step wall-clock and resource envelopes are reported in Supplementary Note 4 (Figs. S1 and S2).

3.3. *ProteoBench benchmark across DIA-NN versions and instruments*

To test whether `quantmsdiann` preserves DIA-NN’s desktop performance under SDRF-driven orchestration and distributed runs, we processed the four public ProteoBench DIA modules (Orbitrap Astral, *Module 7*; single-cell Astral, *Module 9*, PXD049412; timsTOF diaPASEF, *Module 5*, PXD062685; SCIEX ZenoTOF 7600, *Module 10*, PXD070049) and compared the per-precursor identification counts against the ProteoBench public-submission snapshot (29 May 2026) [10]. All runs were library-free (the library predicted in-silico from the FASTA; `quantmsdiann` also supports user-supplied libraries), so for a like-for-like comparison we kept only predicted-from-FASTA DIA-NN community submissions. We report the oldest supported release (1.8.1) and the current 2.5.1-enterprise build, at DIA-NN’s recommended per-version q-values under a uniform 1% global precursor cut-off (Fig. 2); the academic 2.5.1 build and the full per-version, per-module breakdown are in Supplementary Note 5 (Fig. S3).

On the two modules with predicted-library community submissions (Module 7, $n = 15$; Module 5, $n = 8$; Modules 9 and 10 had none), `quantmsdiann` recovers the expected HYE per-species $\log_2$ fold-changes (Fig. 2b) and falls within the community median quantification-error $|\epsilon|$ distribution, at or below its median (Fig. 2c). This holds independent of release: the 1.8.1 and 2.5.1-enterprise builds overlies one another and both sit inside the community cloud, the version-to-version shift in median $|\epsilon|$ (≈ 0.5 –1% in linear fold-change) being far smaller than the community spread. For all the Proteobench datasets explored 2.5.1-enterprise build outperformed 1.8.1 in every benchmark, with no loss of accuracy. Precursors identified in at least one run rose by 15–23% on all four modules (largest on the timsTOF diaPASEF module, 97,149 $\rightarrow$ 119,726), per-run protein groups by 10–14%, and proteins quantified in *every* run by 12–20% (up to +20% on the single-cell Astral module, 6,124 $\rightarrow$ 7,349). This uniform gain, in line with the trend the broader ProteoBench community reports across DIA-NN releases [10], is what makes reanalysing public deposits with the current build worthwhile (Sections 3.4–3.5).

3.4. *Single-cell DIA reanalysis improves with current DIA-NN releases*

To assess `quantmsdiann` on low-input data, where library-free sensitivity and match-between-runs matter most, we reanalysed two label-free single-cell HeLa cohorts, Orbitrap Astral single-cell (PXD046357) and One-Tip (PXD044991), through the workflow under DIA-NN 1.8.1 and the current 2.5.1-enterprise build, both parametrised from the same SDRF (Fig. 3). The newer 2.5.1-enterprise version increased total identifications on both cohorts: on HeLa Astral, union precursors rose from 19,674 to 24,111 (+23%) and protein groups from 3,903 to 4,574 (+17%); on One-Tip, precursors rose from 11,618 to 16,534 (+42%) and protein groups from 1,597 to 2,306 (+44%; Fig. 3a), with the gain holding per cell (Fig. 3b). The improvement is largest in the data-completeness profile

(the number of protein groups quantified across an increasing fraction of cells; Fig. 3c), consistent with stronger cross-run propagation, while quantitative precision and dynamic range are preserved (Fig. 3d,e). As an orthogonal, multiplexed test we reanalysed the first single-cell plexDIA cohort processed through **quantmsdiann** (MSV000093870, Galatidou *et al.* 2024 [18]; mTRAQ 3-channel, Q Exactive), quantifying 2,904 protein groups from the same raw data versus 2,122 in the original analysis (+37%). It recovered 94% of the originally reported groups (1,999 of 2,122) and a further 905 groups at comparable per-cell median depth (1,736 versus 1,784, across 107 and 97 cells), with a QC-matched per-oocyte comparison confirming equivalent single-cell depth (Pearson $r = 0.98$; Fig. 3f and Supplementary Note 6).

3.5. Pan-cohort reanalysis of public cell-line DIA deposits

To show the value of running one SDRF-driven workflow across many independent deposits, we reanalysed five public cell-line DIA panels in a single **quantmsdiann** invocation: NCI-60 (PXD003539, Guo *et al.* 2019 [19]; originally OpenSWATH), ProCan-DepMapSanger (PXD030304, Gonçalves *et al.* 2022 [20]; originally DIA-NN 1.7.x), the Sun *et al.* 2023 PCT-SWATH breast-cancer panel (PXD004701, 76 lines [21]), the MultiPro batch-effect testbed (PXD041421, Wang *et al.* 2023 [22]), and Tognetti *et al.* 2021 (PXD017199, 67 breast-cancer lines [23]) (Fig. 4). For the two cohorts with a published DIA protein-group headline, the reanalysis recovered more protein groups under the original studies' ≥ 2 -unique-peptide criterion, applied identically to both sides: NCI-60 6,328 versus 4,284 (+48%) and ProCan 8,166 versus 6,698 (+22%; Fig. 4a). This ≥ 2 -peptide gain exceeds the change in the total protein-group count: current DIA-NN releases identify more peptides per protein, so proteins that the original analyses supported with a single peptide now clear the stricter two-peptide threshold, expanding the high-confidence protein set rather than only the marginal one. The effect is largest for the older OpenSWATH-based NCI-60 analysis (+48%) and remains substantial for the DIA-NN 1.7-era ProCan panel (+22%). The Sun panel, already analysed with a modern DIA engine, also gained at 1% global FDR (7,412 versus 6,091 protein groups); PXD041421 and PXD017199 carry no published DIA headline and contribute reanalysis-only counts. Across the five cohorts the reanalysis spans 12,845 unique UniProt accessions, of which 5,495 (42.8%) are detected in all five (a pan-cohort core proteome) and a further 1,526 (11.9%) in exactly four. Tissue-level coverage spans 28 pan-cancer tissue categories (Fig. 4b), which maps both the cell lines contributing to each tissue and the union of unique proteins detected per tissue, led by lung, breast, and haematopoietic-and-lymphoid tissue. Because every count comes from a single SDRF-parameterised **quantmsdiann** invocation, the resulting protein matrix is queryable from the published QPX archive and can be filtered, joined, or re-grouped by Cellosaurus, NCIt, or Disease Ontology terms without manual harmonisation. Per-cohort completeness, peptide-per-protein distributions, gene-versus-accession overlap and per-cell-line missingness are provided in Supplementary Note 6 (Figs. S4–S12), which also include an NK-cell Fe-NTA phosphoproteomics reanalysis (PXD049692; comparable phosphopeptide-backbone coverage at 1% FDR). A spatial Deep Visual Proteomics cohort of MYCN-amplified neuroblastoma (PXD064049) was likewise reanalysed and matched the original DIA-NN 1.8.1 analysis at the precursor level (16,443 versus 16,518).

4. Discussion

quantmsdiann automates the path from a deposited DIA dataset to a reproducible quantification table, at a scale that supports systematic reanalysis of public DIA archives. The SDRF-first design encourages experiment annotation upstream of analysis and produces metadata that is

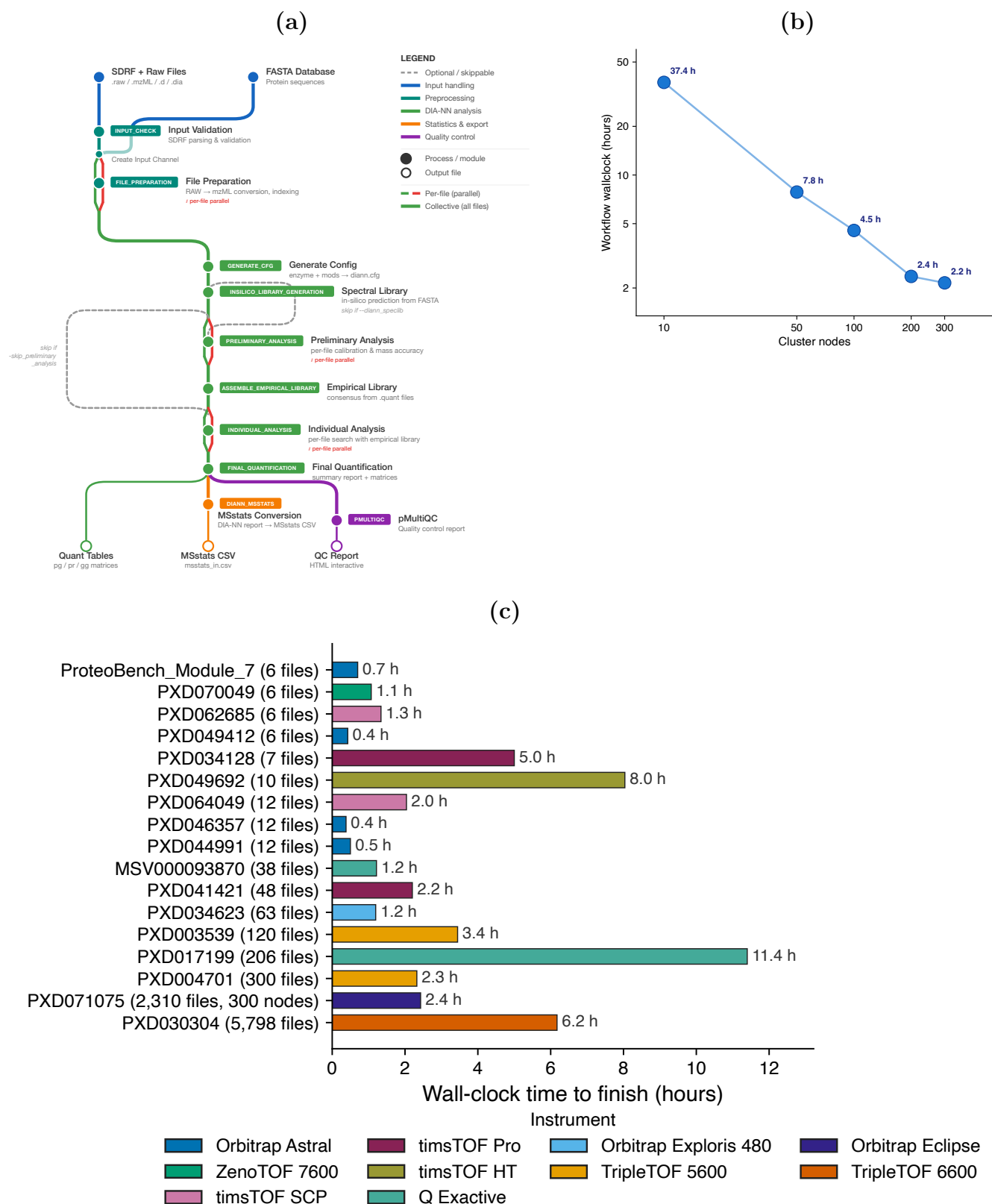


Figure 1: **quantmsdiann pipeline architecture and runtime scaling.** (a) Subway diagram of the **quantmsdiann** workflow: mandatory modules as filled circles, optional/skippable modules as open circles. Per-file parallel stages (red; file preparation, preliminary calibration, individual search) run as one task per raw file, whereas collective stages (green; in-silico library generation, empirical library assembly, final quantification, MSstats conversion, and **pmultiqc** reporting) run once over the full cohort. (b) Workflow wall-clock time versus the number of cluster nodes (Nextflow **executor.queueSize**) on the PXD071075 single-cell cohort. (c) Wall-clock time to finish each reanalysis, one bar per dataset, ordered by cohort size (number of MS data files) and coloured by instrument family. In both (b) and (c) the reported time *excludes* the in-silico spectral-library prediction step (**INSILICO_LIBRARY_GENERATION**), a one-time, cohort-independent cost that can be predicted once and reused, so the bars reflect the per-file quantification time. Times for individual reanalysis of each dataset were obtained from a single analysis of the dataset on a 300-node cluster.

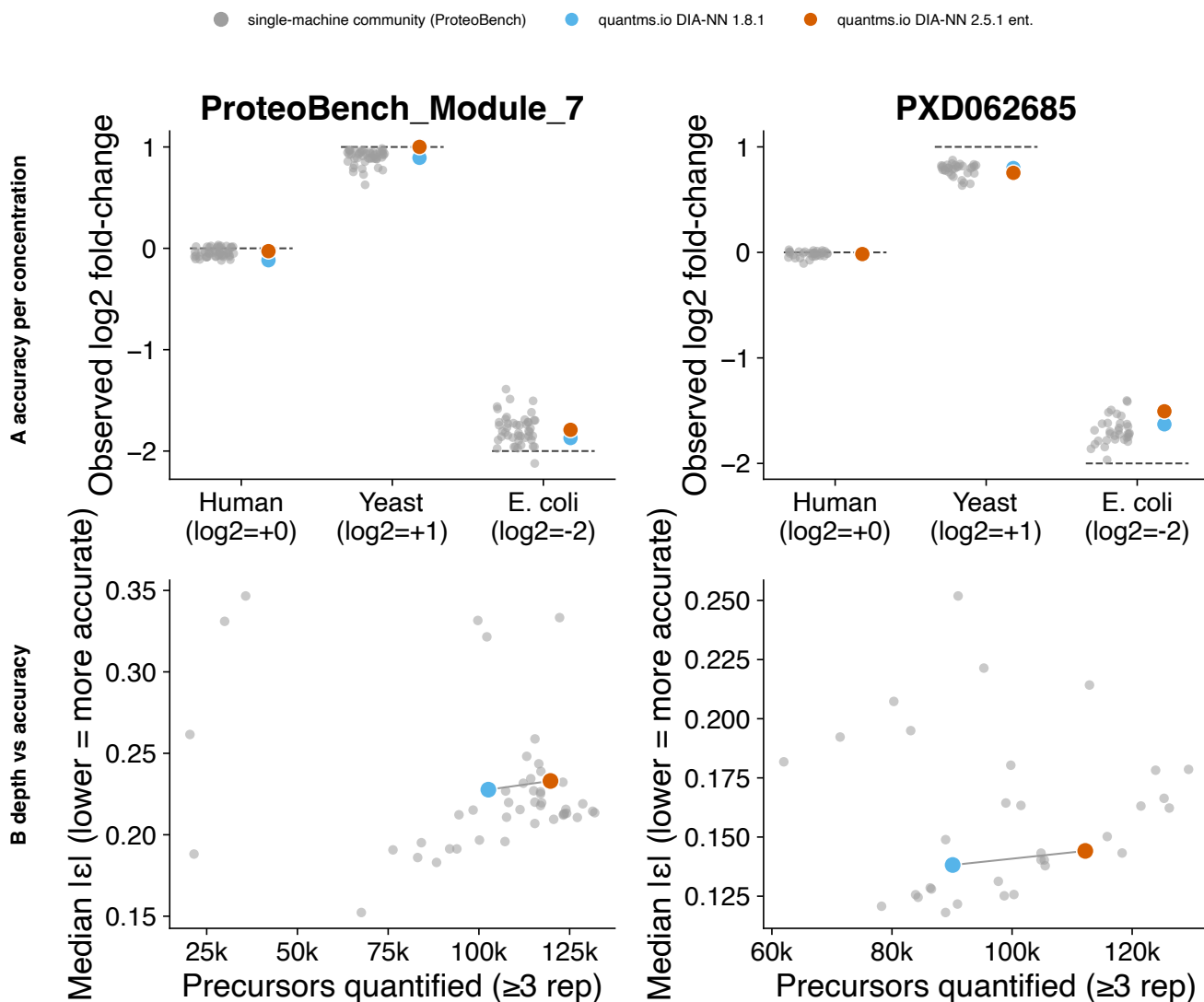


Figure 2: **Distributed quantmsdiann reproduces single-machine results on ProteoBench.** For the two DIA modules with public predicted-from-FASTA community submissions (Orbitrap Astral, Module 7; timsTOF diaPASEF, PXD062685), **quantmsdiann** (DIA-NN 1.8.1 and 2.5.1-enterprise) is compared against the cloud of community submissions, which are single-machine sequential runs. **quantmsdiann** preserves match-between-runs by design: a preliminary pass builds a combined empirical library across all runs, against which each run is re-quantified. (a) Accuracy per concentration: measured $\log_2$ fold-change for each HYE species against the ProteoBench-expected ratio (dashed); community submissions as grey points. (b) Depth versus accuracy: precursors quantified against median- $|e|$; **quantmsdiann** (coloured) sits within the single-machine community cloud (grey), reaching greater depth at comparable accuracy from 1.8.1 to 2.5.1-enterprise. Datasets without predicted-library community comparators (single-cell Module 9/PXD049412; ZenoTOF Module 10/PXD070049) and the per-version protein-group counts are in Supplementary Note 5.

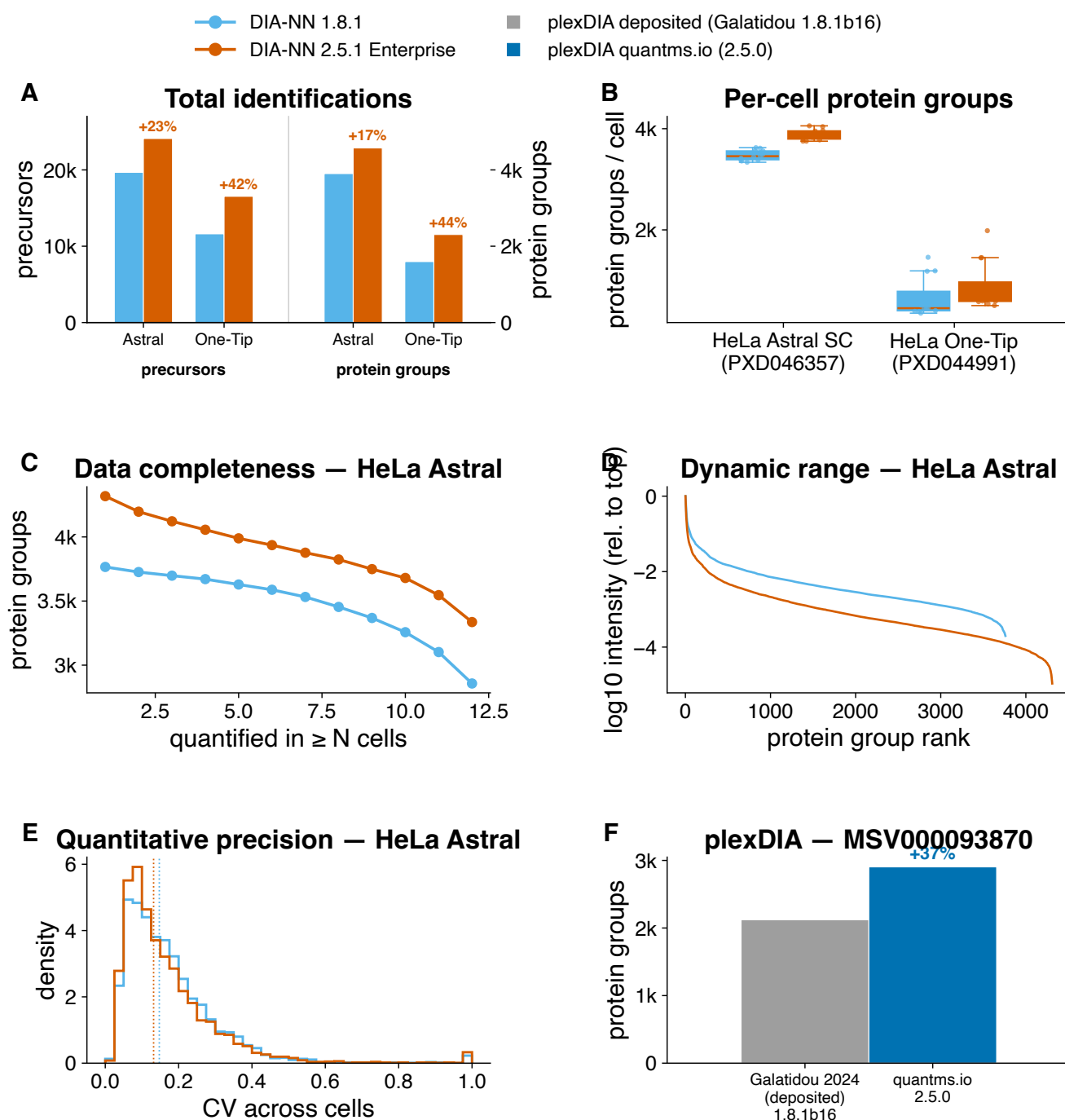


Figure 3: **Single-cell DIA reanalysis by quantmsdiann, comparing DIA-NN 1.8.1 with the current 2.5.1-enterprise build on identical raw data.** Two label-free single-cell HeLa cohorts are shown, Orbitrap Astral single-cell (PXD046357) and One-Tip (PXD044991), alongside an orthogonal multiplexed comparison. (a) Total identifications: union precursors and protein groups across all cells, per build. (b) Protein groups per cell (box and jittered points), showing the newer build lifts depth in every cell. (c) Data-completeness curve (HeLa Astral): protein groups quantified in at least N cells, the match-between-runs signature, where the right endpoint is the complete-profile count. (d) Relative rank-abundance (HeLa Astral), each run normalised to its most-abundant protein, showing extension into lower abundance. (e) Distribution of per-protein coefficient of variation across cells (HeLa Astral), summarising quantitative precision. (f) Multiplexed single-cell plexDIA (MSV000093870, Galatidou 2024; mTRAQ 3-plex): protein groups for the deposited analysis versus the quantmsdiann reanalysis of the same raw data.

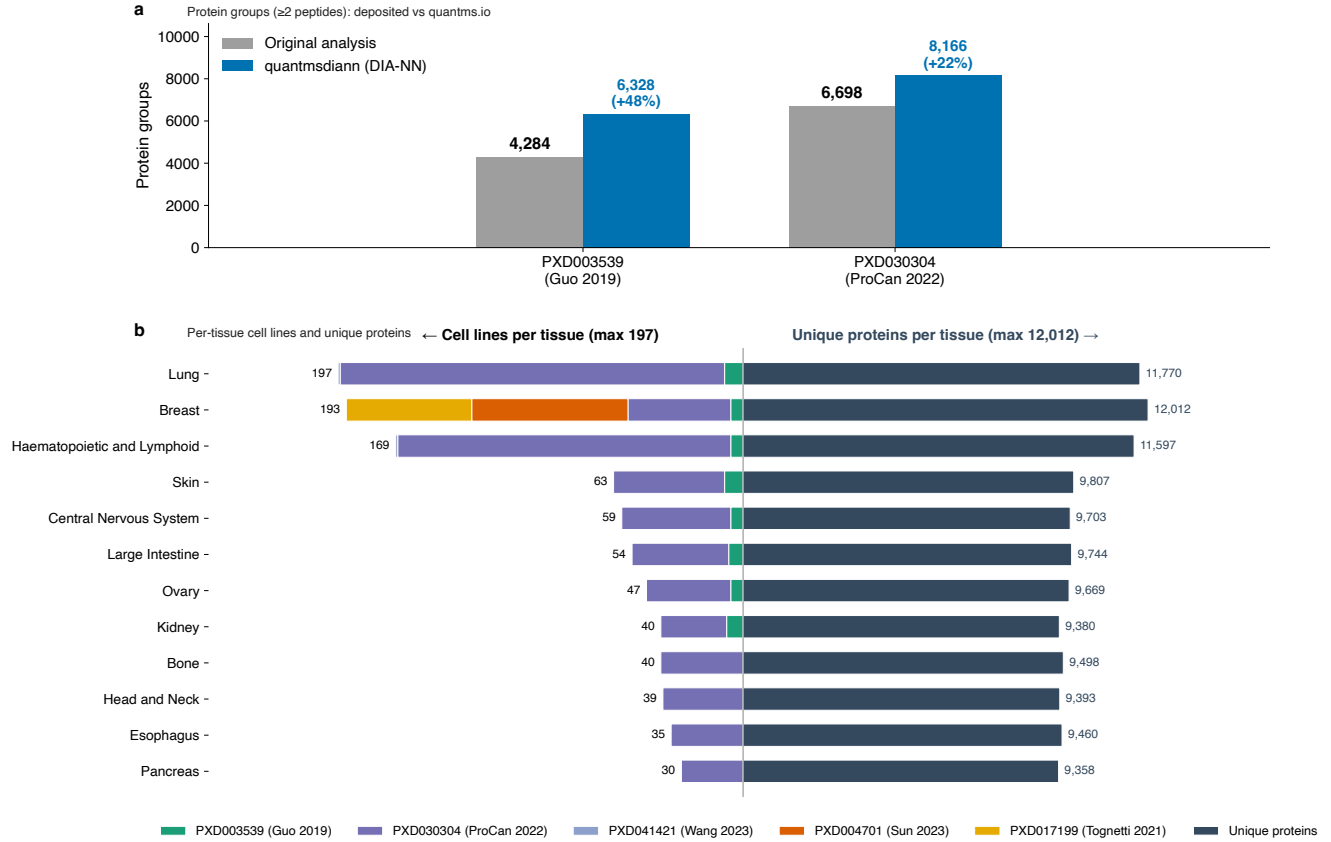


Figure 4: **Pan-cohort of five public DIA cell-line reanalyses by quantmsdiann.** Combines PDX003539 (NCI-60, Guo 2019), PDX030304 (ProCan-DepMapSanger, 949 lines), PDX004701 (Sun 2023, 76 breast cancer lines), PDX041421 (Wang 2023), and PDX017199 (Tognetti 2021, 67 breast cancer lines), reanalysed under a single **quantmsdiann** invocation. **(a)** For the two cohorts with a published DIA protein-group headline (NCI-60 and ProCan), protein groups requiring ≥ 2 unique peptides: the originally published count (grey) and the **quantmsdiann** reanalysis (blue), with the same filter applied to both sides. **(b)** Back-to-back per-tissue summary across the combined panel: cell lines contributing to each tissue, stacked by cohort (left), and the union of unique UniProt proteins detected per tissue (right).

interoperable with the wider proteomics ecosystem [8]; the QPX archive bundles the SDRF, harmonised quantification tables and pipeline provenance into a single Parquet container so that reanalysed results can be archived (e.g. on Zenodo) and re-queried alongside the original raw data. By focusing on DIA-NN rather than the multi-engine scope of the related bigbio/quantms pipeline [7], `quantmsdiann` achieves a leaner dependency surface and faster pipeline-level execution for the DIA-only regime, where the dominant cost is per-file DIA-NN invocation rather than across-engine comparison.

Every reanalysis in this study completed in less than 12h on the EMBL-EBI HPC cluster, independent of cohort size: the 2,300-run single-cell cohort finished in 2.2 h at 300 nodes. This follows from how end-to-end wall-clock is structured, as a largely fixed serial overhead plus the per-file analysis stages. The serial overhead is dominated by in-silico spectral-library generation from the FASTA (the single largest non-parallel step), plus cohort-level consolidation. The per-file stages run as independent tasks across the cluster (Fig. 1b), so their elapsed time grows only sub-linearly with cohort size as files are processed in parallel waves (per-step envelopes in Supplementary Note 4, Fig. S1). Time-to-finish is therefore dominated by the fixed overhead for small cohorts and stays within a few hours even for the largest deposits, so a small dataset is not necessarily faster than a much larger one (Fig. 1c); the serial overhead itself scales with the predicted-library size, growing for wide variable-modification searches such as phosphoproteomics, rather than with the number of files.

Our evaluation covers three complementary aspects of `quantmsdiann`: analytical agreement with the desktop version of DIA-NN (ProteoBench; Fig. 2), recovery against published per-study quantifications (independent reanalyses; Figs. 3, 4), and archive-scale throughput (pan-cohort; Fig. 4). The recovery results suggest that the value of DIA reanalysis is real but time-limited (largest for deposits originally analysed with older or non-DIA-NN engines and shrinking toward parity for recently processed cohorts), a regime an SDRF-driven workflow is suited to address as new DIA-NN versions ship. The workflow already supports broader single-cell and laser-microdissected spatial acquisition modes (Fig. 1a, library-handling branch), and SDRFs for the Brunner [2], Derks [24], Leduc [25], Wang [4], Mund [26] and Makhmut [27] datasets are in preparation.

The cross-DIA-NN-version reproducibility view enabled by `quantmsdiann` is, to our knowledge, the first pipeline-level treatment of this question for DIA-NN. We expect this to become increasingly important as DIA-NN evolves: a deposited single-cell DIA dataset analysed with DIA-NN 1.8 today and re-analysed with DIA-NN 2.x in two years should not produce different biological conclusions for the same data and parameters. In current practice such re-analyses are rarely performed, because the bespoke per-study scripts make them prohibitively expensive, a barrier the single-invocation audit of Section 3.3 removes. By design, `quantmsdiann` ends at harmonised quantification, leaving batch correction, cell-type assignment, and differential-abundance analysis to downstream packages such as `scp` [28] that read its QPX or MSstats output directly; in this sense it complements existing single-cell DIA workflows rather than competing with them. `quantmsdiann` is open-source under the MIT license and is openly developed as a community effort within the nf-core and bigbio proteomics ecosystem, so that it can continue to track new DIA-NN releases and acquisition modes as they appear.

Data Availability

All datasets reanalysed in this study are publicly available through ProteoMeXchange. Cell-line bulk DIA reanalyses: PRIDE PXD003539 (Guo 2019, NCI-60), PXD004701 (Sun 2023), PXD030304 (ProCan-DepMapSanger), PXD017199 (Tognetti 2021), PXD041421 (Wang 2023). Spatial Deep

Visual Proteomics reanalysis: PRIDE PXD064049 (MYCN-amplified neuroblastoma, diaPASEF). Scalability experiment: PRIDE PXD071075 (Wu 2025). ProteoBench benchmarks: PXD049412 (single-cell, Module 9), PXD062685 (diaPASEF, Module 5), PXD070049 (ZenoTOF, Module 10), and Module 7 (Astral) datasets. The full list of PRIDE accessions used in the pan-cohort of DIA cell-line reanalyses is provided in Supplementary Note 1 (Table S1). `quantmsdiann` is available at <https://github.com/bigbio/quantmsdiann> under the MIT license; documentation is hosted at <https://quantmsdiann.quantms.org/>. The pipeline is implemented in Nextflow DSL2 ($\geq 25.10.4$) and packages each step as a versioned Docker, Singularity, or Apptainer container image. DIA-NN 1.8.1 is distributed publicly via BioContainers; because the DIA-NN license restricts redistribution of later releases, containers for DIA-NN 1.9 and above are built locally from the recipes in <https://github.com/bigbio/quantms-containers> (see Experimental Procedures). Pipeline version v2.2.0 (release codename “Chongqing”, 2026-06-16) was used for the analyses reported in this paper. Analysis scripts that generated the figures in this manuscript are available at <https://github.com/bigbio/quantmsdiann-paper> (this repository).

Supplemental Data

This article contains supplemental data. The Supplementary Notes provide the full PRIDE accession list (Supplementary Note 1, Table S1), DIA-NN container-build recipes (Note 2), the filtering and protein-counting rules (Note 3), per-process runtime and resource envelopes (Note 4, Figs. S1–S2), the ProteoBench per-version identification breakdown (Note 5, Fig. S3), per-cohort completeness, peptide-per-protein and gene-versus-accession overlap figures (Note 6, Figs. S4–S12), and a table of software, formats, and public resources with their repositories (Note 7, Table S2).

Conflict of Interest

The authors declare no competing interests.

Acknowledgments

M.B. is supported by the Science and Technology Innovation Key R&D Program of Chongqing (CSTB2023TIAD-STX0002). A.L. acknowledges support from MCIN (Recovery, Transformation and Resilience Plan); the Basque Government “Biotechnology Complementary Plan Applied to Health” funded by the European Union NextGeneration EU (PRTR-C17.I1; PRTR-C17.I01.P01.S13) (AAAA_ACG_AY_2539/22_05); and an EMBO Scientific Exchange Grant (number 10015/2023). Y.P.-R. acknowledges support from the Wellcome Trust [223745/Z/21/Z], BBSRC [BB/V018779/1, BB/X001911/1], EPSRC [EP/Y035984/1], a UKRI single-cell proteomics (SCP) grant [UKRI701], and EMBL core funding.

Author Contributions

QXY, YS and CD developed the pipeline. HW helped build the containers and contributed to the workflow implementation and manuscript writing. AL annotated the SDRF metadata and helped run the workflow. TS, FC, NS and JN assisted with the DIA-NN reanalyses and manuscript writing; TS additionally contributed to the workflow design, and JN tested the tool and identified multiple issues. OK provided feedback on the plexDIA datasets. VD provided DIA-NN expertise and guidance on cross-version reproducibility. MB contributed to manuscript writing and supervised the students; MB and YPR conceived and supervised the project. QXY and YPR wrote the original draft, with input from all authors. All authors read and approved the final manuscript.

Abbreviations

CLI	command-line interface
DDA	data-dependent acquisition
DIA	data-independent acquisition
DIA-NN	data-independent acquisition neural networks (DIA search and quantification software)
diaPASEF	data-independent acquisition with parallel accumulation–serial fragmentation
FDR	false discovery rate
HPC	high-performance computing
HYE	human, yeast, and <i>E. coli</i> (ProteoBench three-proteome mixture)
IQR	interquartile range
LSF	Load Sharing Facility (HPC scheduler)
MS	mass spectrometry
MS1/MS2	first-/second-stage mass spectrometry (precursor/fragment scans)
PASEF	parallel accumulation–serial fragmentation
QPX	Quantitative Proteomics eXchange
SDRF	Sample and Data Relationship Format

References

- [1] V. Demichev, C. B. Messner, S. I. Vernardis, K. S. Lilley, M. Ralser, DIA-NN: neural networks and interference correction enable deep proteome coverage in high throughput, *Nat Methods* 17 (2020) 41–44. doi:10.1038/s41592-019-0638-x.
- [2] A.-D. Brunner, M. Thielert, C. Vasilopoulou, C. Ammar, F. Coscia, A. Mund, et al., Ultra-high sensitivity mass spectrometry quantifies single-cell proteome changes upon perturbation, *Mol Syst Biol* 18 (2022) e10798. doi:10.15252/msb.202110798.
- [3] F. Meier, A.-D. Brunner, M. Frank, A. Ha, I. Bludau, E. Voytik, et al., diaPASEF: parallel accumulation-serial fragmentation combined with data-independent acquisition, *Nat Methods* 17 (2020) 1229–1236. doi:10.1038/s41592-020-00998-0.
- [4] J. Wang, Y. Huang, F. Lu, Q. Xu, Z. Yang, Y. Jiang, et al., Benchmarking informatics workflows for data-independent acquisition single-cell proteomics, *Nat Commun* 16 (2025) 10276. doi:10.1038/s41467-025-65174-4.
- [5] Y. Perez-Riverol, C. Bandla, D. J. Kundu, S. Kamatchinathan, J. Bai, S. Hewapathirana, N. S. John, A. Prakash, M. Walzer, S. Wang, J. A. Vizcaíno, et al., The PRIDE database at 20 years: 2025 update, *Nucleic Acids Res* 53 (D1) (2025) D543–D553. doi:10.1093/nar/gkae1011.
- [6] E. W. Deutsch, N. Bandeira, Y. Perez-Riverol, V. Sharma, J. J. Carver, L. Mendoza, D. J. Kundu, C. Bandla, S. Kamatchinathan, S. Hewapathirana, et al., The ProteomeXchange consortium in 2026: making proteomics data FAIR, *Nucleic Acids Res* 54 (D1) (2026) D459–D469. doi:10.1093/nar/gkaf1146.
- [7] C. Dai, J. Pfeuffer, H. Wang, P. Zheng, L. Käll, T. Sachsenberg, et al., quantms: a cloud-based pipeline for quantitative proteomics enables the reanalysis of public proteomics data, *Nat Methods* 21 (2024) 1603–1607. doi:10.1038/s41592-024-02343-1.

- [8] C. Dai, A. Füllgrabe, J. Pfeuffer, E. M. Solovyeva, J. Deng, P. Moreno, et al., A proteomics sample metadata representation for multiomics integration and big data analysis, *Nat Commun* 12 (2021) 5854. doi:10.1038/s41467-021-26111-3.
- [9] P. A. Ewels, A. Peltzer, S. Fillinger, H. Patel, J. Alneberg, A. Wilm, et al., The nf-core framework for community-curated bioinformatics pipelines, *Nat Biotechnol* 38 (2020) 276–278. doi:10.1038/s41587-020-0439-x.
- [10] R. Devreese, C. Jachmann, B. Van Puyvelde, H. Webel, W. Bittremieux, C. Chiva, et al., ProteoBench: the community-curated platform for comparing proteomics data analysis workflows, *bioRxivPreprint* (2025). doi:10.64898/2025.12.09.692895.
- [11] UniProt Consortium, UniProt: the universal protein knowledgebase in 2025, *Nucleic Acids Res* 53 (2025) D609–D617. doi:10.1093/nar/gkae1010.
- [12] N. Hulstaert, J. Shofstahl, T. Sachsenberg, M. Walzer, H. Barsnes, L. Martens, et al., ThermoRawFileParser: modular, scalable, and cross-platform RAW file conversion, *J Proteome Res* 19 (2020) 537–542. doi:10.1021/acs.jproteome.9b00328.
- [13] S. Kamatchinathan, S. Hewapathirana, C. Bandla, S. Insua, J. A. Vizcaíno, Y. Perez-Riverol, pridepy: a Python package to download and search data from PRIDE database, *J Open Source Softw* 10 (107) (2025) 7563. doi:10.21105/joss.07563.
- [14] F. da Veiga Leprevost, B. A. Grüning, S. Alves Aflitos, H. L. Röst, J. Uszkoreit, H. Barsnes, M. Vaudel, et al., BioContainers: an open-source and community-driven framework for software standardization, *Bioinformatics* 33 (16) (2017) 2580–2582. doi:10.1093/bioinformatics/btx192.
- [15] Q.-X. Yue, C. Dai, S. Kamatchinathan, A. Larrea, Y. Perez-Riverol, et al., pmultiqc: an open-source, lightweight, and metadata-oriented QC reporting library for MS proteomics, *bioRxivPreprint* (2025). doi:10.1101/2025.11.02.685980.
- [16] P. Ewels, M. Magnusson, S. Lundin, M. Källér, MultiQC: summarize analysis results for multiple tools and samples in a single report, *Bioinformatics* 32 (2016) 3047–3048. doi:10.1093/bioinformatics/btw354.
- [17] I. Virshup, D. Bredikhin, L. Heumos, G. Palla, G. Sturm, A. Gayoso, et al., The scverse project provides a computational ecosystem for single-cell omics data analysis, *Nat Biotechnol* 41 (2023) 604–606. doi:10.1038/s41587-023-01733-8.
- [18] S. Galatidou, A. A. Petelski, A. Pujol, K. Lattes, L. B. Latorraca, T. Fair, M. Popovic, R. Vassena, N. Slavov, M. Barragán, Single-cell proteomics reveals decreased abundance of proteostasis and meiosis proteins in advanced maternal age oocytes, *Mol Hum Reprod* 30 (7) (2024) gaae023. doi:10.1093/molehr/gaae023.
- [19] T. Guo, A. Luna, V. N. Rajapakse, C. C. Koh, Z. Wu, W. Liu, Y. Sun, H. Gao, M. P. Menden, C. Xu, et al., Quantitative proteome landscape of the NCI-60 cancer cell lines, *iScience* 21 (2019) 664–680. doi:10.1016/j.isci.2019.10.059.
- [20] E. Gonçalves, R. C. Poulos, Z. Cai, S. Barthorpe, S. S. Manda, N. Lucas, A. Beck, et al., Pan-cancer proteomic map of 949 human cell lines, *Cancer Cell* 40 (2022) 835–849.e8. doi:10.1016/j.ccell.2022.06.010.

- [21] R. Sun, W. Ge, Y. Zhu, A. Sayad, A. Luna, M. Lyu, S. Liang, et al., Proteomic dynamics of breast cancer cell lines identifies potential therapeutic protein targets, *Mol Cell Proteomics* 22 (2023) 100602. doi:10.1016/j.mcpro.2023.100602.
- [22] H. Wang, K. P. Lim, W. Kong, H. Gao, B. J. H. Wong, S. X. Phua, T. Guo, W. W. B. Goh, MultiPro: DDA-PASEF and diaPASEF acquired cell line proteomic datasets with deliberate batch effects, *Sci Data* 10 (2023) 858. doi:10.1038/s41597-023-02779-8.
- [23] M. Tognetti, A. Gabor, M. Yang, V. Cappelletti, J. Windhager, O. M. Rueda, et al., Deciphering the signaling network of breast cancer improves drug sensitivity prediction, *Cell Syst* 12 (5) (2021) 401–418.e12. doi:10.1016/j.cels.2021.04.002.
- [24] J. Derks, A. Leduc, G. Wallmann, R. G. Huffman, M. Willetts, S. Khan, et al., Increasing the throughput of sensitive proteomics by plexdia, *Nat Biotechnol* 41 (2023) 50–59. doi:10.1038/s41587-022-01389-w.
- [25] A. Leduc, L. Khoury, J. Cantlon, S. Khan, N. Slavov, Massively parallel sample preparation for multiplexed single-cell proteomics using npop, *Nat Protoc* 19 (2024) 3750–3776. doi:10.1038/s41596-024-01033-8.
- [26] A. Mund, F. Coscia, A. Kriston, R. Hollandi, F. Kovács, A.-D. Brunner, et al., Deep visual proteomics defines single-cell identity and heterogeneity in primary human cancers, *Nat Biotechnol* 40 (2022) 1231–1240. doi:10.1038/s41587-022-01302-5.
- [27] A. Makhmut, D. Qin, S. Fritzsche, J. Nimo, J. König, F. Coscia, A framework for ultra-low-input spatial tissue proteomics, *Cell Syst* 14 (2023) 1002–1014.e5. doi:10.1016/j.cels.2023.10.003.
- [28] C. Vanderaa, L. Gatto, The current state of single-cell proteomics data analysis, *Curr Protoc* 3 (2023) e658. doi:10.1002/cpz1.658.